Chapter 18, Problem 55P

Problem

Let $f(t) = 5e{-}(t{-}2)u(t)$. Find $F(\omega)$ and use it to find the total energy in $f(t)$.

Step-by-step solution

Step 1 of 2

$$f(t) = 5e^{-(t-2)}u(t)$$

$$\Rightarrow F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}\,dt$$

$$= \int_{0}^{\infty} 5e^{-(t-2)}e^{-j\omega t}\,dt$$

$$= 5e^2\int_{0}^{\infty} e^{-(1+j\omega)t}\,dt$$

$$= -\frac{5e^2}{1+j\omega}\left|e^{-(1+j\omega)t}\right|_0^{\infty}$$

$$F(\omega) = \frac{5e^2}{1+j\omega}$$

Step 2 of 2

now,

$$W = \frac{1}{\pi}\int_0^{\infty} |F(\omega)|^2\,d\omega \qquad \left[|F(\omega)^2| = F(\omega)*F(\omega) = \frac{25e^4}{1+\omega^2}\right]$$

$$= \frac{1}{\pi}\int_0^{\infty} \frac{25e^4}{1+\omega^2}\,d\omega$$

$$= \frac{25e^4}{\pi}\left[\tan^{-1}\omega\right]_0^{\infty}$$

$$= \frac{25e^4}{\pi}\frac{\pi}{2}$$

$$= \frac{25e^4}{2} \qquad [e = 2.718]$$

$$\boxed{W = 682.5J}$$